

University of Southampton

Faculty of Engineering and Physical Sciences

Electronics and Computer Science

MSc University of Southampton

by

Yufei Tang

December 2023

ELEC6217

Wireless Transceiver Design and Implementation

Assignment 2

1.1 Problem Restatement

Study the "FM Demodulation" example for information flow.

1.2 Finding

We found that the simulated waveform plot of the signal on the LabView is almost 0. This is because everyone's signals are concentrated at the same frequency, so there is too much noise to drown out the real signal. The signal amplitude was improved after modifying the carrier frequency.

2.1 Problem Restatement

Try the "PSK Tx" and "PSK Rx" examples.

2.2 Finding

We used pi/4DQPSK as the modulation method, and chose the other group as the transmitter and us as the receiver. The initial setup is shown in Figure 2.1:

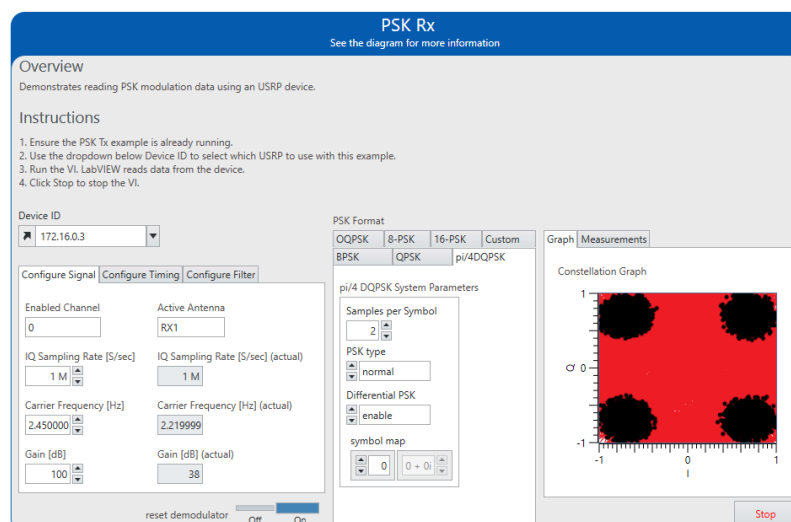


Figure 2.1

Then Carrier Frequency is be changed, which is not matched with the transmitter. And the result can be seen in the Figure 2.2:

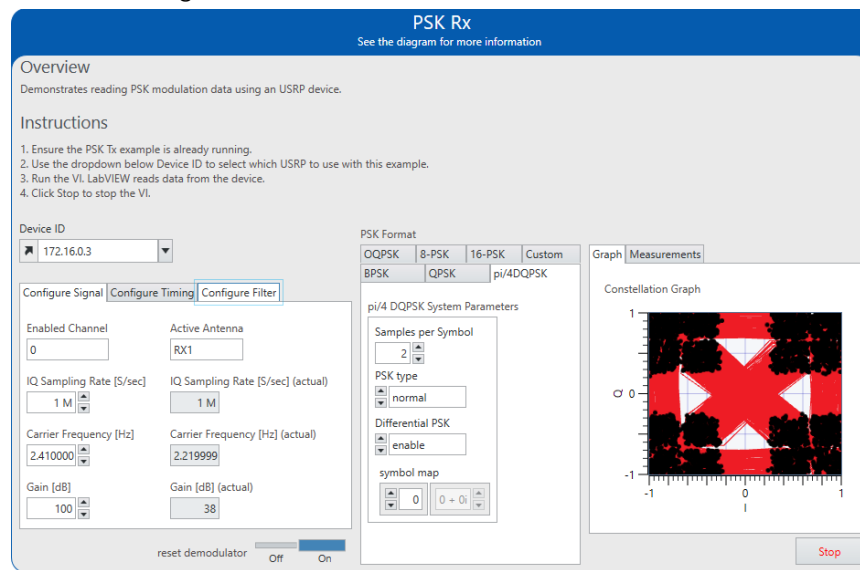


Figure 2.2

Deterioration in the aggregation of signal values in the constellation charts can be found in the above figure. And then the upsampling rate is changed from 2 to 16, the result is shown in the Figure 2.3.

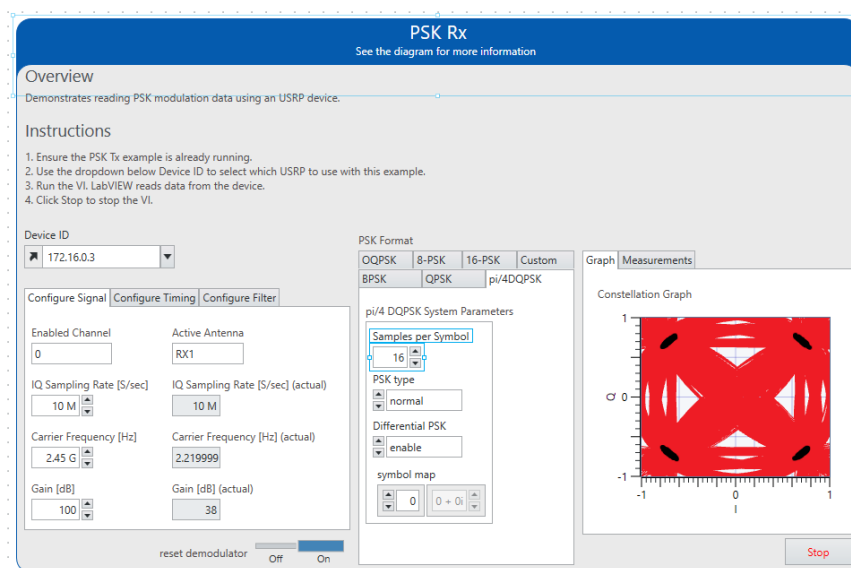


Figure 2.3

The constellation map of the signals became much clearer.

Then the Gain is be set to 20, as you can see in the chart below the constellation diagram of the signals has become blurred again.

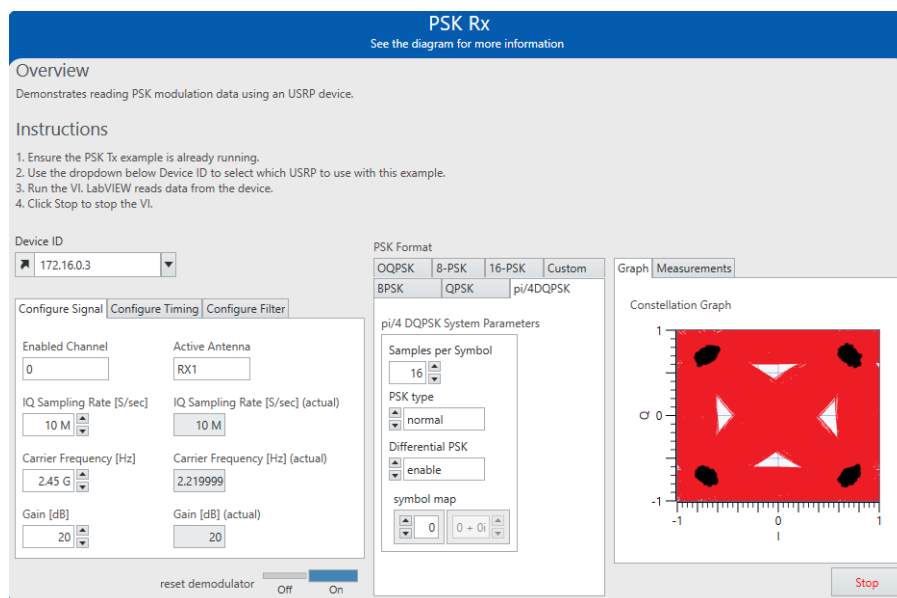
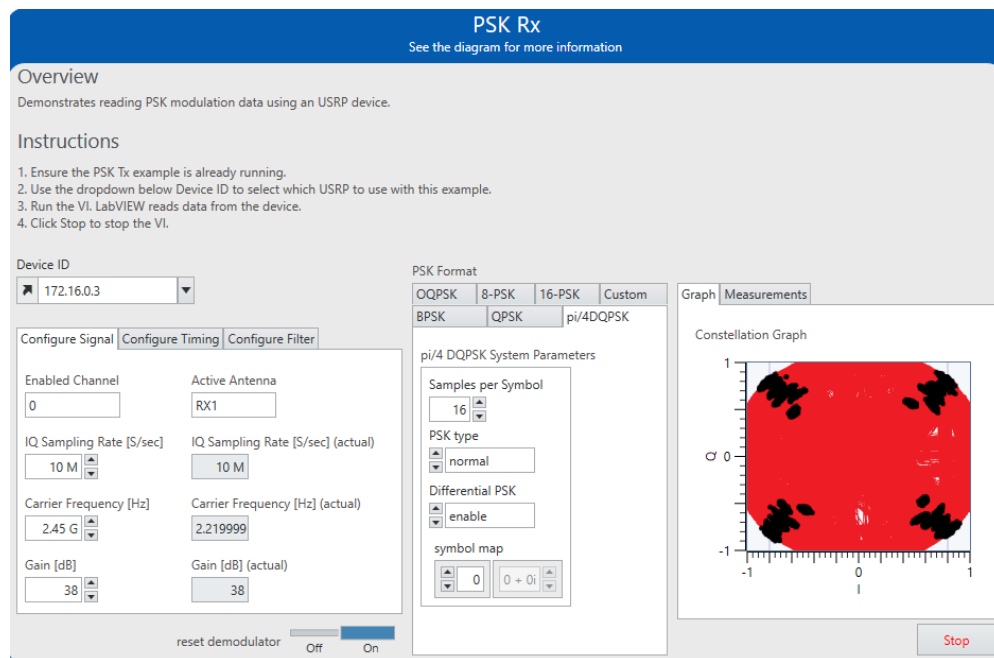


Figure 2.4

Then the type of the filter is to be changed from the Root Rise Cosine Filter to the Rise Cosine Filter, and the result is shown in the Figure 2.5.



As shown above, the constellation Graph presents a serious error.

3. Conclusion

For the constellation diagram to be rendered correctly, the sender and receiver must be made to match. Each time after modifying the parameter values in the main page you need to go into the subgraph to see if the status of the corresponding part is updated, otherwise the experiment may fail because the data is not updated.